

EGO / HUMANGUARD

Project: Sovereign Edge-AI Safety & Navigation Platform

Status: Production-Ready / Phase 7 Completed

1. System Architecture Blueprint

The following structural schema demonstrates the end-to-end data flow of the Ego platform, illustrating strict local isolation, zero-cloud dependency, and secure optional synchronization.



2. Deep-Dive Core Layers & Execution Mechanics

1. Field Ingestion & Camera Binding

Components: CameraStreamBinding

Ingests raw video feeds and edge sensors, applying smart frame-skipping and capping maxFramesPerSecond to protect memory buffers and prevent GC spikes on constrained hardware.

2. Vision & Hazard Engine

Components: EgoHazardCaptureEngine

Performs local computer vision processing (zero-cloud). Implements strict deduplication and cooldown intervals (minCaptureInterval) to eliminate redundant log writes and storage bloat.

3. Secure Local Storage & Encrypted Blackbox

Components: EgoDatabase / EgoTelemetryService

The security core. Stores all GPS coordinates, acceleration telemetry, and incident snapshots inside an encrypted local database operating completely offline for absolute data sovereignty.

4. Dual-Mode HUD & Local Navigation

Components: Ego HUD (Pedestrian & Operator Modes)

An ultra-smooth, zero-jank UI completely isolated from heavy processing threads via background workers. Serves dual purposes: safe personal pedestrian routing and real-time industrial machinery telemetry.

5. Secure Local-to-Remote Sovereign Sync Engine

Components: SecureBlackboxSyncEngine

Packages pending blackbox logs into lightweight batches when connectivity is available, utilizing exponential backoff retries. Local records are strictly preserved until a verified server acknowledgment (HTTP ACK) is received.

3. Strategic Value & Regulatory Compliance

- **Absolute Privacy & Legal Compliance:** Operates 100% on-device without continuous raw video streaming, adhering strictly to global and regional data privacy laws, including Morocco's **CNDP Law 09-08** and Europe's **GDPR**.
- **Edge-Optimized Performance:** Engineered for constrained hardware environments (such as NVIDIA Jetson Orin/ Nano and Raspberry Pi 5) with minimal thermal output and optimized memory usage.
- **Legal & Operational Resilience:** Encrypted blackbox recording combined with fault-tolerant background synchronization guarantees zero data loss during power loss or network failures.

The Ask & Strategic Partnership Goals

We are seeking hardware development partners, automotive/industrial manufacturers, and acceleration programs to support our **Pilot Field Deployment Phase**. Specifically, we aim to secure hardware testing environments on dedicated chips (e.g., NVIDIA Jetson Orin, NXP i.MX8, or industrial ARM platforms) and establish commercial pilot trials with fleet operators and smart industrial facilities.

Executive Summary

Ego / HumanGuard bridges the gap between high-performance edge computing and real-world industrial/personal safety. With all seven development phases fully validated and optimized, the platform is fully prepared for field deployment, hardware integration, and commercial scaling.